2027 Planting Plan

16 open spots, 9 cultivars: 13 apples, 1 pear, 1 cherry, 1 peach. Recommended cultivars for the open 2027 spots; none are purchased yet.

> **What this is:** the recommended cultivar for each spot left open for 2027, chosen for a short-season U-pick orchard at about 4,750 ft: the three Urban-apple spots, the dead pear, cherry, and peach spots, and the empty apple positions at the east end of R6. Every cultivar links to its profile in the cultivar guide, with sources.
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> **Status:** recommended, not bought, not planted. Chosen so far: Harrow Delight pear, Black Gold cherry on Mazzard, Intrepid peach, and the apple mix below; the trees will be bought rather than grafted. The planned trees are not counted as pollen partners on the pollination grid until they are in the ground.

Buy list

Cultivar	Trees and spots	Ripens	Rootstock	Pollination	Why, and cautions
Apple: Pristine	1: R6-C9	Early	Semi-dwarf wanted; nurseries list M.27 or EMLA 26 dwarfs, too small here	Flowering group 2, diploid	One-tree test of a very early yellow apple. Sources disagree on ripening (mid-July or August) and zone (4 to 9 or 5 to 8), and Purdue rates it susceptible to cedar-apple rust, with junipers on site. No source gave its flowering group, so group 2 is assumed.
Apple: Williams' Pride	3: R6-C7, R6-C8, R5-C9	Early	Semi-dwarf wanted; G.214 is cold-hardy and fire-blight resistant	Flowering group 2, diploid	Early block: the first apples of the season. Ripens unevenly over about a week, so plan on more than one picking; keeps about six weeks in the fridge. Sources put its bloom early, in group 1 to 2, so group 2 is a planning value.
Apple: Dayton	2: R6-C4, R6-C11	Mid	Semi-dwarf wanted; nurseries list EMLA 27 or M.26 dwarfs, too small for 18 x 20 ft spacing	Flowering group 3, diploid	Large crisp sweet-tart apple for mid to late September. Cummins lists bloom group 3 and diploid. Raintree puts harvest in late September but Cummins calls it early-ripening, so its timing here is uncertain.
Apple: Priscilla	2: R5-C10, R6-C12	Mid	Semi-dwarf wanted; G.890 (about 60 percent of standard) is offered	Flowering group 3, diploid	Early-to-mid September apple; hardy to about -50 F. Medium fruit with a mild flavor and a licorice note; bears heavily, so thin the crop.
Apple: Freedom	3: R6-C5, R6-C6, R6-C13	Late	Size-controlling semi-dwarf wanted; nurseries often ship it on MM.111, which stays large	Flowering group 3, diploid	Large striped-red apple for early October; very disease resistant. Ripens with Delicious (about Oct 5 at Geneva, NY), so it is a gamble in a short season. Vigorous, spreading tree. No source gave its flowering group, so group 3 is assumed.
Apple: Haralson	2: R4-C9, R6-C10	Late	M.7 semi-dwarf; B.9 dwarf is also offered but too small here	Flowering group 4, diploid	October pie and cider apple; zone 3, late bloom. Roots to Fruits lists bloom group 4; an earlier group-3 claim could not be confirmed. Ripens in mid-October, the latest of the new apples, and is tart rather than a fresh-eating favorite.
Pear: Harrow Delight	1: R8-C1	Early	OHxF 87 (semi-dwarf, cold-hardy, fire-blight resistant)	Needs a partner (not self-fertile)	Fire-blight-resistant European pear that ripens early. Not self-fertile. Only one pollination partner counts within 50 ft (Red Bartlett at 49 ft), so fruit set may be light. A fifth Asian pear would have Hosui and Hardy Giant beside it.
Cherry: Black Gold	1: R9-C2	Mid	Mazzard or another canker-tolerant stock; avoid Gisela 6, the most canker-susceptible (Intermountain Fruit)	Self-fertile	Hardy, late-blooming, canker-resistant dark sweet cherry. Blooms later than Bing, so it adds no pollen for the Bing or Black Tartarian trees, which still depend on Lapins at R10-C2. Birds need netting.
Peach: Intrepid	1: R11-C10	Mid	Krymsk 86 was suggested (cold-hardy, anchors well, near-standard size); canker tolerance of the rootstock is unverified	Self-fertile	Bud-hardy, self-fertile mid-season freestone. Peaches here carry bacterial canker (untreated positives). Prune only in dry weather, use dormant copper, and buy clean certified stock.

16 trees in all (13 apples, 1 pear, 1 cherry, 1 peach). Ask nurseries for semi-dwarf trees (about 60 to 75 percent of standard; G.890, G.214 or M.7 for apples). The grid is 18 x 20 ft and the site is dry and windy, so dwarf stocks (M.27, M.9, B.9) would need permanent stakes and leave gaps. Confirm that each nursery ships the item to Idaho before ordering; some retailers exclude Idaho for some items.

Where they go

Draft placement, rows listed south to north as in the planting map. C1 is the west end of each row and C13 the east end by the driveway; every other row is staggered half a spot. Where it works, same-cultivar trees sit side by side for signage and easy picking: the Williams' Pride trio and two of the three Freedom. The other cultivars are split between the west and east halves, and the east end (C11 to C13) gets three different group-3 apples because it has the fewest pollen partners.

Row	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13
R4	CO	UNK	HC	GD	UNK	ZT	WC	UNK	**HAL**	UNK	AF	HR	DLG
R5	CAM	ID	HG	LB	SS	CL	MT	ZT	**WLP**	**PSC**	HR	CC	KER
R6	KER	GS	RS	**DAY**	**FRD**	**FRD**	**WLP**	**WLP**	**PTN**	**HAL**	**DAY**	**PSC**	**FRD**
R8	**HDL**	HA	HG	HHC	BT	MW	AU	HHC	SD	AU	BT	CE	—
R9	BB	**BGD**	BB	ST	NKI	SWT	HG	NKI	—	—	—	—	—
R11	MP	UNK	GT	—	—	—	IPL	CR	RH	**INT**	FT	CH	RC

☐ planned 2027 cultivar code ⬚ a spot held open for a dead or relocated tree ☐ tree already on hand, by its code. Codes: PTN = Pristine; WLP = Williams' Pride; DAY = Dayton; PSC = Priscilla; FRD = Freedom; HAL = Haralson; HDL = Harrow Delight; BGD = Black Gold; INT = Intrepid.

Pollination check

Using the pollination grid's rules: partners are other cultivars within 50 ft that bloom in the same or an adjacent period and are compatible (apples: diploid donors whose flowering group is within one of the tree's own; cherries and pears: the bloom charts on the grid page). Living trees and the other planned trees count.

Spot	Cultivar	Partners within 50 ft	Nearest partners
R4-C9	Haralson	8	Aztec Fuji (22 ft), GoldRush (22 ft), Priscilla (22 ft)
R5-C9	Williams' Pride	8	Zestar (18 ft), Priscilla (18 ft), Pristine (22 ft)
R5-C10	Priscilla	10	Harrison (18 ft), Williams' Pride (18 ft), Haralson (22 ft)
R6-C4	Dayton	9	Freedom (18 ft), Sweet Sixteen (22 ft), Cortland (34 ft)
R6-C5	Freedom	7	Dayton (18 ft), Cortland (22 ft), Sweet Sixteen (22 ft)
R6-C6	Freedom	6	Williams' Pride (18 ft), Cortland (22 ft), Macintosh (22 ft)
R6-C7	Williams' Pride	8	Freedom (18 ft), Macintosh (22 ft), Zestar (22 ft)
R6-C8	Williams' Pride	8	Pristine (18 ft), Zestar (22 ft), Macintosh (34 ft)
R6-C9	Pristine	7	Williams' Pride (18 ft), Priscilla (22 ft), Harrison (34 ft)
R6-C10	Haralson	5	Dayton (18 ft), Harrison (22 ft), Priscilla (22 ft)
R6-C11	Dayton	8	Haralson (18 ft), Priscilla (18 ft), CrimsonCrisp (22 ft)
R6-C12	Priscilla	6	Dayton (18 ft), Freedom (18 ft), CrimsonCrisp (22 ft)
R6-C13	Freedom	4	Priscilla (18 ft), CrimsonCrisp (34 ft), Dayton (36 ft)
R8-C1	Harrow Delight	1	Red Bartlett (49 ft)
R9-C2	Black Gold	Self-fertile; none needed	None counted
R11-C10	Intrepid	Self-fertile; none needed	None counted

Every planned apple has at least 4 partners. Black Gold is self-fertile and blooms later than the Bing and Black Tartarian trees beside it, so it adds no pollen for them; they still depend on Lapins. Planned trees never change the partners shown for the trees already on the pollination grid.

Season fit

Only 4 of the 51 apples you already have ripen early, and 24 ripen late or very late, which is a gamble with 60 to 100 frost-free days. The new apples add 4 early, 4 mid-season, and 5 late trees; the late ones (Freedom and Haralson) are the riskiest of the new picks, so the plan adds no more.

Apples	Early	Mid	Late	Very late	Not identified	Total
Living trees now	4	17	14	10	6	51
New in 2027	4	4	5	0	0	13
After the plan	8	21	19	10	6	64

Bands are relative to a cool northern apple harvest: Early before about Sept 10, Mid about Sept 10 to Oct 5, Late about Oct 5 to 25, Very late after Oct 25 (see the guide). Unidentified apples are not banded.

Why these picks

The site

- About 4,750 ft, nearly flat; Simonton loam with soil pH about 6.2; 12 to 16 in of precipitation, with July and August almost dry (see the water-harvesting plan).
- NRCS lists only **60 to 100 frost-free days** for this soil's setting. The four NOAA stations serving Elmore County share an 80-day season, and the longest single station allows 160 days; the nearest similar-elevation station (Hill City, 5,100 ft) has a median season of 111 days. Best estimate here: 70 to 110 days and USDA zone 5a to 5b. Sources disagree and the USDA map was not checked for this exact point, so treat these as estimates, not readings.
- In general, bloom-frost risk runs worst to best from apricot to peach, sweet cherry, pear, and apple (a general horticultural ranking, not something measured here). That is why apples carry the plan and the three replacements are chosen for hardiness: late bloom or frost-tolerant buds.

What a U-pick orchard needs

- Fruit people eat off the tree, a harvest that stretches over weeks (the orchard had four early apples), trees small enough to pick without ladders, and few sprays. That points to semi-dwarf trees and disease-resistant cultivars.
- Sweet cherries and European pears are the harder U-pick crops (birds and splitting; pears are picked hard and ripen off the tree), so the single replacement trees are a modest bet, not a block.

Bacterial canker in the stone fruit

> The inventory flags 24 stone-fruit records as Pseudomonas (bacterial canker) positive and untreated: 10 cherries, 5 apricots, 9 peaches. The dead Royalton cherry and J.H. Hale peach were among them. That is a pattern, not proof of why they died; the causes of all three losses are unknown.
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> So the replacements are chosen and handled with canker in mind: a cherry with reported canker resistance on Mazzard, buying clean certified stock, pruning only in dry weather, and dormant copper sprays at budbreak and leaf drop (Intermountain Tree Fruit Production Guide). Cold-air drainage, winter injury, and drought stress all make canker worse; see the water-harvesting plan for how the stone-fruit rows drain.

Suggestions not followed

- **Encore peach** topped the first round of research for being the latest ripener. It ripens in fall, after Elberta, which is already a gamble here, so Intrepid replaces it.
- **Gisela 6 cherry rootstock** was suggested for Black Gold. The Intermountain Tree Fruit Production Guide lists Gisela 6 as the most bacterial-canker-susceptible rootstock and Mazzard as the least, so the plan says Mazzard.

Alternates and what to avoid

Spot	Picked	Alternates	Avoid
13 apple spots	The six-cultivar mix above	Redfree (August, very rust resistant, but soft and the only confirmed source was Canadian). Honeygold is a poor fit: fire blight very susceptible and rust susceptible, with junipers on site.	Braeburn: susceptible to scab, fire blight, rust, and mildew, and needs a long warm fall (one Braeburn is already in the orchard plan at R2-C2).
R8-C1 pear	Harrow Delight	Potomac; Magness (pollen-sterile, so it needs a partner and pollinates nothing)	Pineapple again (the first one died; cause unknown)
R9-C2 cherry	Black Gold on Mazzard	WhiteGold (self-fertile, zone 5 to 7, resists brown rot and canker; blooms earlier than Black Gold, so it would add a second pollen source for the Bing trees); Skeena	Royalton again (it died); Gisela 6 rootstock
R11-C10 peach	Intrepid	Madison (late August, hardy buds)	Encore (ripens too late); J.H. Hale again (it died)

Not checked yet: extra Honeycrisp or Zestar for name recognition, Spartan, SnowSweet, and tart cherries and plums for the other empty positions and R13 to R15.

Ripening watch: apples you already have

These apples ripen very late, after about Oct 25 in a cool northern climate. With 70 to 110 frost-free days they may not ripen reliably here. Watch them for the first few crops; any that cannot ripen before the hard freeze can be top-worked (grafted over) to an early or mid-season cultivar.

Cultivar	Trees	Ripening, from its profile
Arkansas Black	1: R2-C10	Very late (Early Nov per Raintree; about Oct 20 near Madison, Wisconsin per Lapacek's Orchard)
Aztec Fuji	2: R3-C6, R3-C9	Very late (Late October)
Campfield	1: R5-C1	Very late (Early Nov per Raintree; Raintree and Cummins both say it ripens late)

Cultivar	Trees	Ripening, from its profile
Fuji	1: R2-C4	Very late (Oct to Nov; one of the last apples. Needs a very long season, so it may not fully ripen in cold, short-season sites.)
GoldRush	1: R3-C10	Very late (Late Oct to early Nov in upstate New York; fruit hangs on well after it is ripe)
Granny Smith	1: R6-C2	Very late (Very late per Orange Pippin Trees; needs a long, warm season. WSU lists it as mid-season for Washington.)
Newtown Pippin	1: R3-C13	Very late (Late October in the eastern US (Orange Pippin); ripens poorly without a hot summer and autumn)
Pink Lady (Cripps Pink)	2: R2-C7, R3-C5	Very late (Late October in Willis's listing; needs roughly 150 frost-free days (Pomiferous))

Braeburn at R2-C2 belongs on this watch list too: its band is Late, but it is disease-susceptible and wants a long warm fall. Golden Delicious, Idared, Red Delicious, and Mutsu are less likely to be a problem: the Intermountain Tree Fruit Production Guide's apple table, written for Utah, lists them as cold hardy with September to October harvests. Prairie has a shorter season than most of Utah, so treat that as encouraging, not proof.

Sources and caveats

- Site: NRCS Web Soil Survey report (soil pH, frost-free period); US Growers Guide: Planting zones in Elmore County, Idaho (zones 5a to 7a; last spring freeze May 1 to June 16 and first fall freeze Sept 4 to Oct 8 across four NOAA stations); Bield: Elmore County frost dates (Hill City 1 W, 5,100 ft: median last frost May 30, first fall frost Sept 18, 111 days, zone 5b); University of Idaho Extension: backyard tree fruit in harsher climates (apples, apricots, cherries, pears, and plums can be grown in short-season zones with the right cultivars).
- Cherries: Oregon State Extension: sweet cherry compatibility and bloom timing chart (Black Gold in pollination period IV; Bing and Stella in II; Lapins in I); Intermountain Tree Fruit Production Guide: bacterial canker (rootstocks and copper sprays).
- Apples: Intermountain Tree Fruit Production Guide: apple varieties for Utah (harvest times and hardiness for Golden Delicious, Idared, Red Delicious, Mutsu, and Haralson).
- Cultivar facts, ratings, and ripening dates are in each profile's sources in the cultivar guide. The flowering groups for Williams' Pride, Freedom, and Pristine are planning values that no source confirmed; Dayton's and Priscilla's come from Cummins, and sources disagree on Haralson's (Roots to Fruits says group 4). Sources also disagree on Pristine's ripening, zone, and rust rating. The buy list says so.
- Confirm that each nursery ships the item to Idaho and offers the rootstock you want before ordering. This plan does not pick nurseries.